

Game Theory Twelve

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1 Activation Function Switching

1.1 Two-Player Two-Strategy Example

To determine the effect that different activation functions have on BNN dynamics, I revisited the two-player game with payoff matrices

$$A = \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 7 & 5 \\ 6 & 8 \end{pmatrix}.$$

In this game, neither player possesses a strictly dominant strategy, leading to multiple Jacobian switching points. As demonstrated in previous weeks, this reduced the error order from the expected fourth order of RK4 to second order.

I investigated alternative activation functions for the BNN dynamics, out of a desire to find a smooth alternative to ReLU. In particular, I examined these three activation functions:

1. $\varphi(x) = \ln(1 + e^x)$ (Soft ReLU)
2. $\varphi(x) = \ln(1 + e^x) - \ln(2)$
3. $\varphi(x) = \ln(1 + xe^x)$

The latter two activation functions increased convergence time, ruling them out as viable replacements for ReLU. Soft ReLU actually had faster convergence with ReLU and because of the Nash Equilibrium and matrix construction, had the Nash Equilibrium as a fixed point. That said, I will investigate whether soft ReLU continues to produce more efficient BNN dynamics.

A further topic of investigation is activation function engineering, to increase convergence rates while ensuring that each Nash equilibrium is represented in the set of solutions across randomized initializations.

2 Three-Strategy BNN Dynamics

Next, I looked at whether BNN dynamics would converge to Nash equilibrium in the two-player, three strategy case. For this investigation, I used the payoff matrices

$$A = \begin{pmatrix} 0 & 3 & 0 \\ 1 & 0 & 4 \\ 2 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 0 & 1 \\ 0 & 1 & 4 \end{pmatrix}.$$

There are three Nash equilibria for this game:

$$p^{1*} = (1, 0, 0), \quad p^{2*} = (0, 1, 0).$$

$$p^{1*} = \left(0, \frac{2}{3}, \frac{1}{3}\right), \quad p^{2*} = \left(\frac{4}{5}, 0, \frac{1}{5}\right).$$

$$p^{1*} = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right), \quad p^{2*} = \left(\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\right).$$

Numerical experiments showed that, under BNN dynamics, the strategies of both players converged either towards the first or second Nash equilibrium. Even when the strategy profiles of both players were initiated close to the third Nash equilibrium, their strategies did not converge there. I will further investigate determining whether a Nash equilibrium is an unstable equilibrium.

Finally, I recovered similar error order results in the three strategy case as with the two strategy case. In both cases, when the Jacobian of the BNN dynamics remains consistent, the error order of the RK4 method is the expected fourth order. When the strategies of both players travel across multiple regions with different Jacobians, this is reduced to second order.

3 Finite Difference Methods

The second portion of my presentation reviewed topics from LeVeque's *Finite Difference Methods for Ordinary and Partial Differential Equations*. This textbook, specifically Chapters 5.3, 5.4, 7.2, and 7.3, was used in Gianluca's candidacy exam as required knowledge for numerical methods.

3.1 Finite Difference Approximations

Using Taylor expansions, the first derivative can be approximated by

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}.$$

The second derivative can be approximated by

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}.$$

More generally, higher-order derivatives can be approximated by solving linear systems involving Taylor expansions.

3.2 Boundary Value Problems

Consider the heat equation

$$u_t = \kappa u_{xx}.$$

Boundary value problems specify conditions on the spatial boundary, while initial value problems specify conditions at the initial time. Common boundary conditions include:

Dirichlet Boundary Conditions

$$u(a, t) = \alpha(t), \quad u(b, t) = \beta(t).$$

Neumann Boundary Conditions

$$u_x(a, t) = \alpha(t), \quad u_x(b, t) = \beta(t).$$

Robin Boundary Conditions

$$u(a, t) + c_1 u_x(a, t) = g_1(t), \quad u(b, t) + c_2 u_x(b, t) = g_2(t).$$

3.3 Local and Global Error

Let

$$U^{n+1} = \mathcal{A}(U^n)$$

be a numerical scheme.

Define the local truncation error $\mathcal{A}(U) - \mathcal{A}(u)$ by τ .

The global error is

$$E = -\mathcal{A}^{-1}\tau.$$

We say the scheme is:

- **Consistent** if $\lim_{h \rightarrow 0} \|\mathcal{A}^{-1}\| < \infty$
- **Stable** if $\lim_{h \rightarrow 0} \|\tau\| = 0$
- **Convergent** if $\lim_{h \rightarrow 0} \|E\| = 0$

The Lax Equivalence Principle states:

$$\text{Consistency} + \text{Stability} \implies \text{Convergence}.$$

4 LeVeque 5.3 Numerical Methods for ODEs

4.1 Forward Euler

$$u^{n+1} = u^n + hf(t_n, u^n).$$

4.2 Backward Euler

$$u^{n+1} = u^n + hf(t_{n+1}, u^{n+1}).$$

4.3 Trapezoid Method

$$u^{n+1} = u^n + \frac{h}{2} [f(t_n, u^n) + f(t_{n+1}, u^{n+1})].$$

4.4 Multistep Methods

Leapfrog Method

$$u^{n+1} = u^{n-1} + 2hf(t_n, u^n).$$

Backward Differentiation Formula

$$\sum_{j=0}^k \alpha_j u^{n+j} = h\beta_k f(t_{n+k}, u^{n+k}).$$

5 LeVeque 5.4 Truncation Error

Taylor expansions can be substituted into numerical methods to determine the local truncation error. For example, the Leapfrog method has local truncation error

$$\tau^n = \mathcal{O}(h^2).$$

6 LeVeque 7.2 and 7.3 Absolute Stability

Given the ODE

$$u' = \lambda u,$$

numerical stability depends on whether

$$z = h\lambda$$

lies inside the stability region of the numerical method.

6.1 Forward Euler Stability Region

Forward Euler gives

$$u^{n+1} = (1 + z)u^n.$$

Thus the stability region is

$$|1 + z| \leq 1.$$

6.2 Trapezoid Stability Region

For the trapezoid method,

$$u^{n+1} = \frac{1 + z/2}{1 - z/2} u^n.$$

The stability region becomes

$$\left| \frac{1 + z/2}{1 - z/2} \right| \leq 1.$$

7 Probability Topics

7.1 Dr. Rizzolo Topics

- Brownian motion
- Stochastic differential equations
- Fokker–Planck equations

Primary references include:

- Karatzas and Shreve: *Brownian Motion and Stochastic Calculus*
 - This reference dependent upon review
- Pavliotis: *Stochastic Processes and Applications*

8 Fluid Dynamics Topics

8.1 Dr. Guyenne Topics

- Derivation of Navier–Stokes equations
- Compressible versus incompressible flow
- Potential flow
- Hamiltonian systems
- Numerical methods for Hamiltonian systems

Primary reference:

- Landau and Lifshitz: *Fluid Mechanics*

For Hamiltonian systems, I am considering using reference material regarding HJB equations, contingent upon Dr. Guyenne’s approval.